

CS 3006

Parallel and Distributed Computing

Lecture 8

Danyal Farhat
FAST School of Computing
NUCES Lahore

Mapping Techniques

Lecture's Agenda

- **Mapping Schemes**
 - **Static Mapping**
 - **Dynamic Mapping**
- **Schemes for Static Mapping**
 - **Block distribution (1D, 2D)**
 - **Cyclic distribution**
 - **Block-Cyclic distribution**
 - **Randomized-Block distribution (solution: 1D)**
 - **Randomized-Block distribution (solution: 2D)**

Lecture's Agenda (Cont.)

- **Random Distribution**
- **Graph Partitioning**
- **Summary**
- **Additional Resources**

Mapping Schemes

Static Mapping:

- Distributing the tasks among the processes **before execution** of the program
 - E.g., usually used in situation where **total number of tasks and their sizes** are known before the execution of the program
- Easy to implement in message passing paradigm

Dynamic Mapping:

- When total number of tasks **are not known** a priori
 - (OR) when task sizes are unknown
 - ✓ In this case static mapping can lead to **serious load-imbalances**

Schemes for Static Task-Process Mapping

Array Distribution Schemes

- In a decomposition based on partitioning data, **mapping** the relevant **data** onto the processes is equivalent to **mapping tasks** onto processes
- Commonly used array mapping schemes:
 - Block distribution
 - ✓ 1D and 2D
 - Cyclic and block-cyclic distribution

Schemes for Static Mapping

- **Block distribution (1D)**

row-wise distribution

P_0
P_1
P_2
P_3
P_4
P_5
P_6
P_7

column-wise distribution

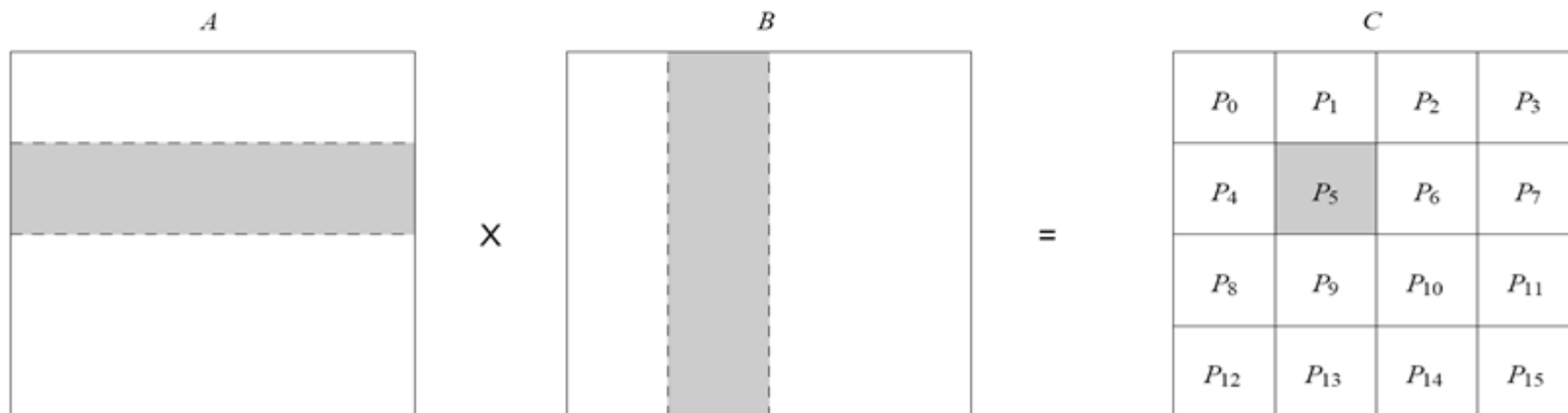
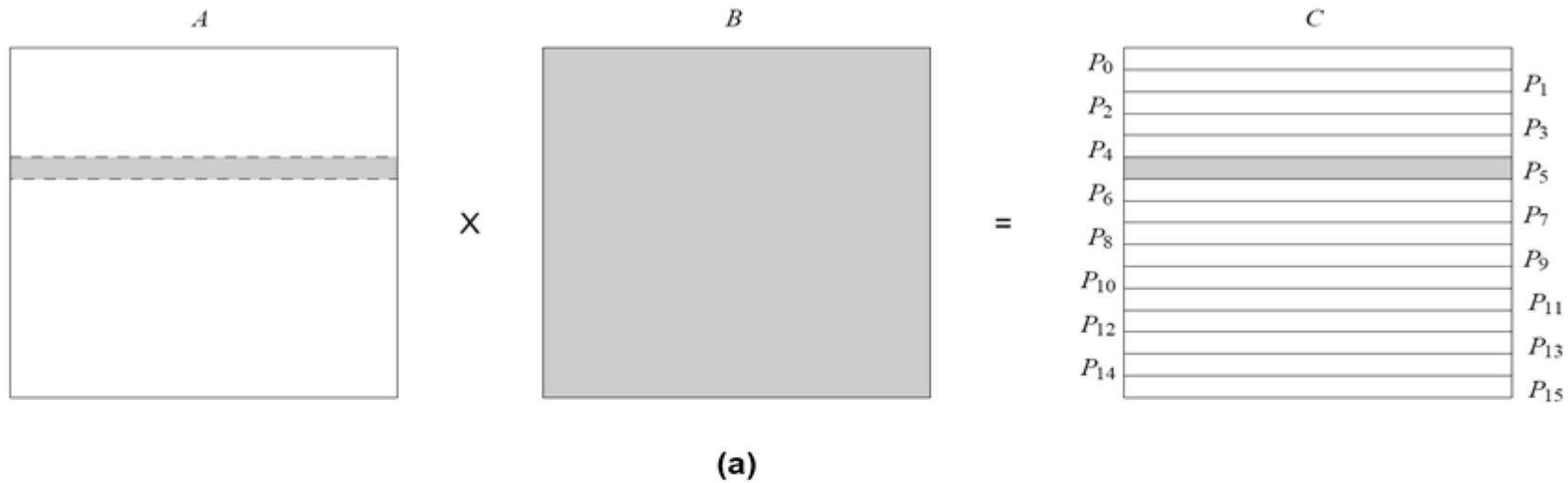
P_0	P_1	P_2	P_3	P_4	P_5	P_6	P_7
-------	-------	-------	-------	-------	-------	-------	-------

- **Block distribution (2D)**

P_0	P_1	P_2	P_3
P_4	P_5	P_6	P_7
P_8	P_9	P_{10}	P_{11}
P_{12}	P_{13}	P_{14}	P_{15}

P_0	P_1	P_2	P_3	P_4	P_5	P_6	P_7
P_8	P_9	P_{10}	P_{11}	P_{12}	P_{13}	P_{14}	P_{15}

Schemes for Static Mapping (Cont.)



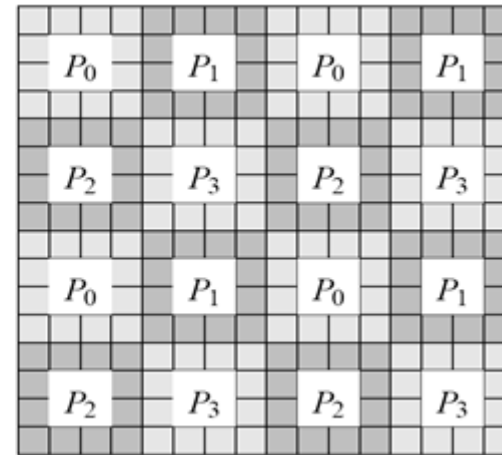
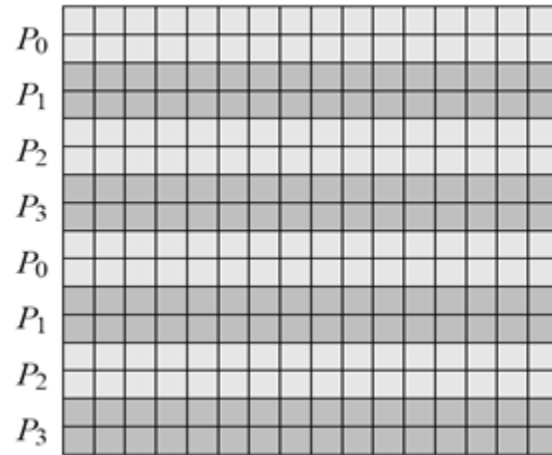
Schemes for Static Mapping (Cont.)

- Cyclic distribution (HERE array size=4 x 4 and p=3)

P0	P1	P2	P0
P1	P2	P0	P1
P2	P0	P1	P2
P0	P1	P2	P0

Schemes for Static Mapping (Cont.)

- **Block-Cyclic distribution**



Schemes for Static Mapping (Cont.)

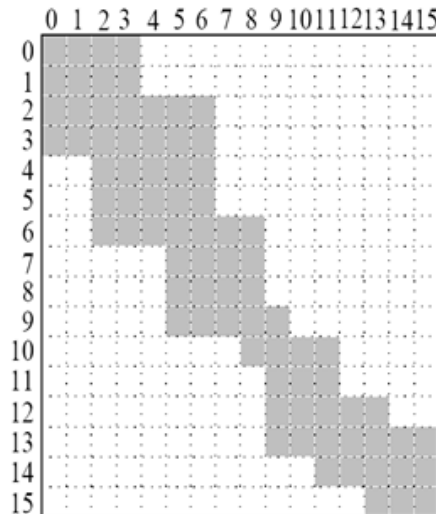
- Randomized-Block distribution (solution: 1D)

$V = [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]$

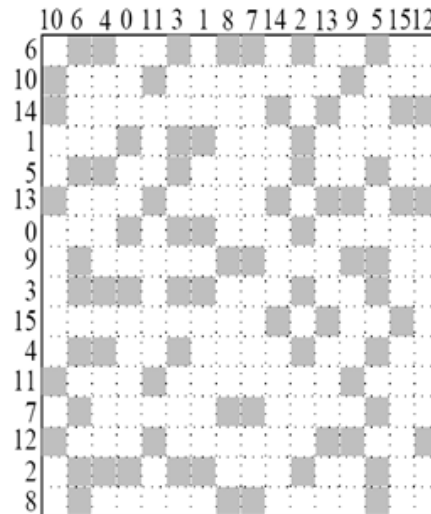
$\text{random}(V) = [8, 2, 6, 0, 3, 7, 11, 1, 9, 5, 4, 10]$

mapping = 8 2 6 0 3 7 11 1 9 5 4 10
 └─┘ └─┘ └─┘ └─┘
 P_0 P_1 P_2 P_3

- Randomized-Block distribution (solution: 2D)



(a)



(b)

P_0	P_1	P_2	P_3
P_4	P_5	P_6	P_7
P_8	P_9	P_{10}	P_{11}
P_{12}	P_{13}	P_{14}	P_{15}

(c)

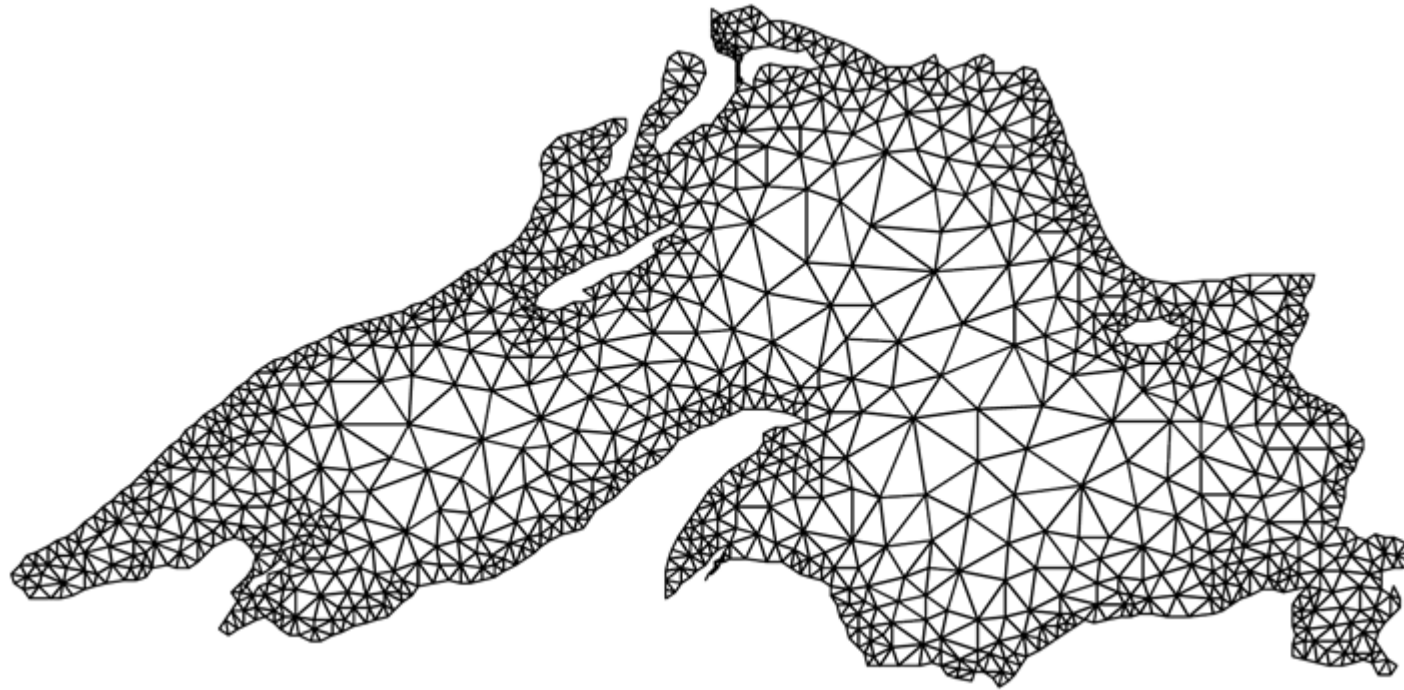
Schemes for Static Mapping (Cont.)

Why randomized block cyclic distribution is not always used?

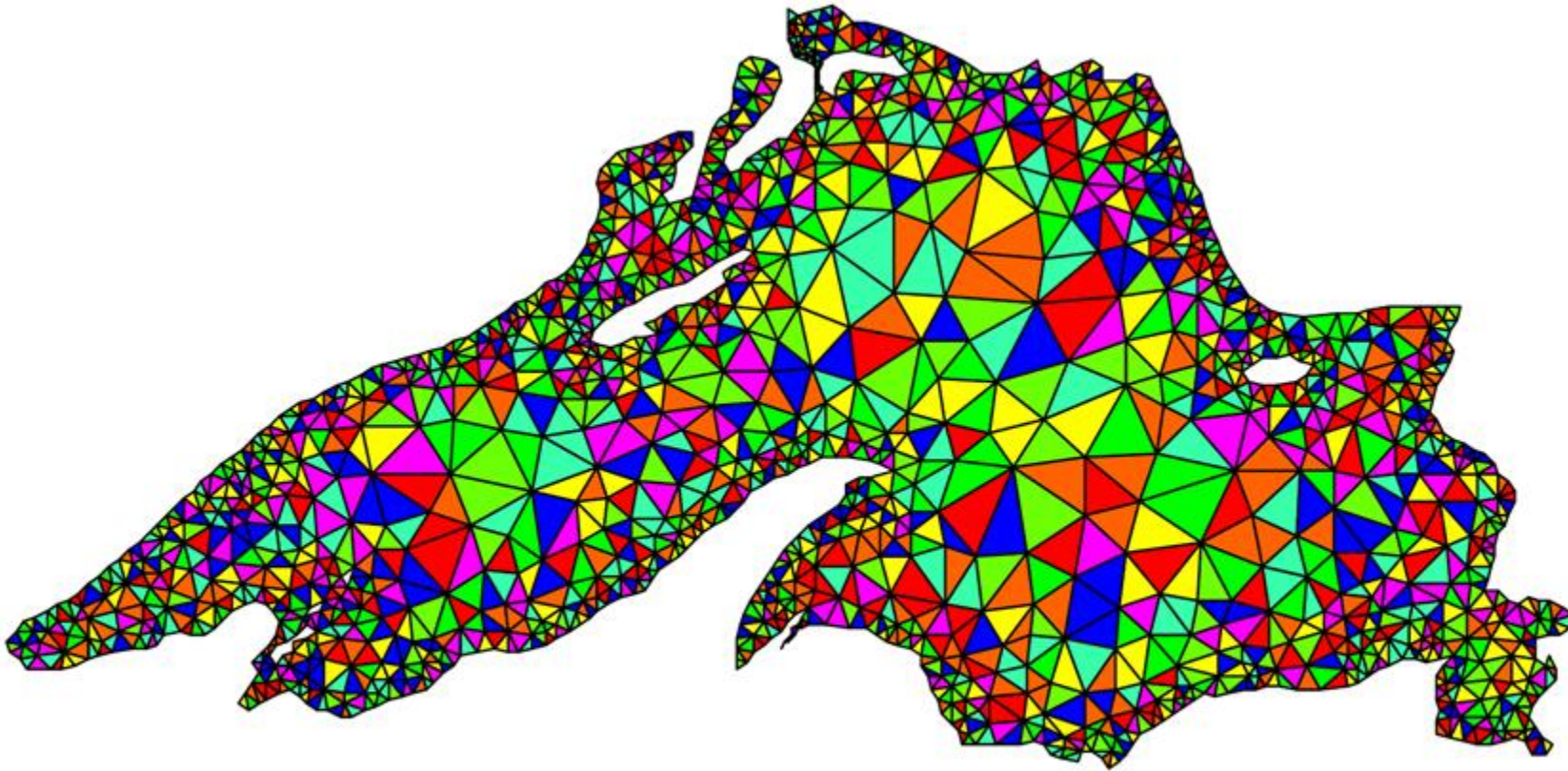
- **Poor spatial locality**
- **High inter-process interaction**

Example

- A **simulation model** (using a mesh of tasks) for finding dispersion of water contaminant in a lake at different intervals of time.

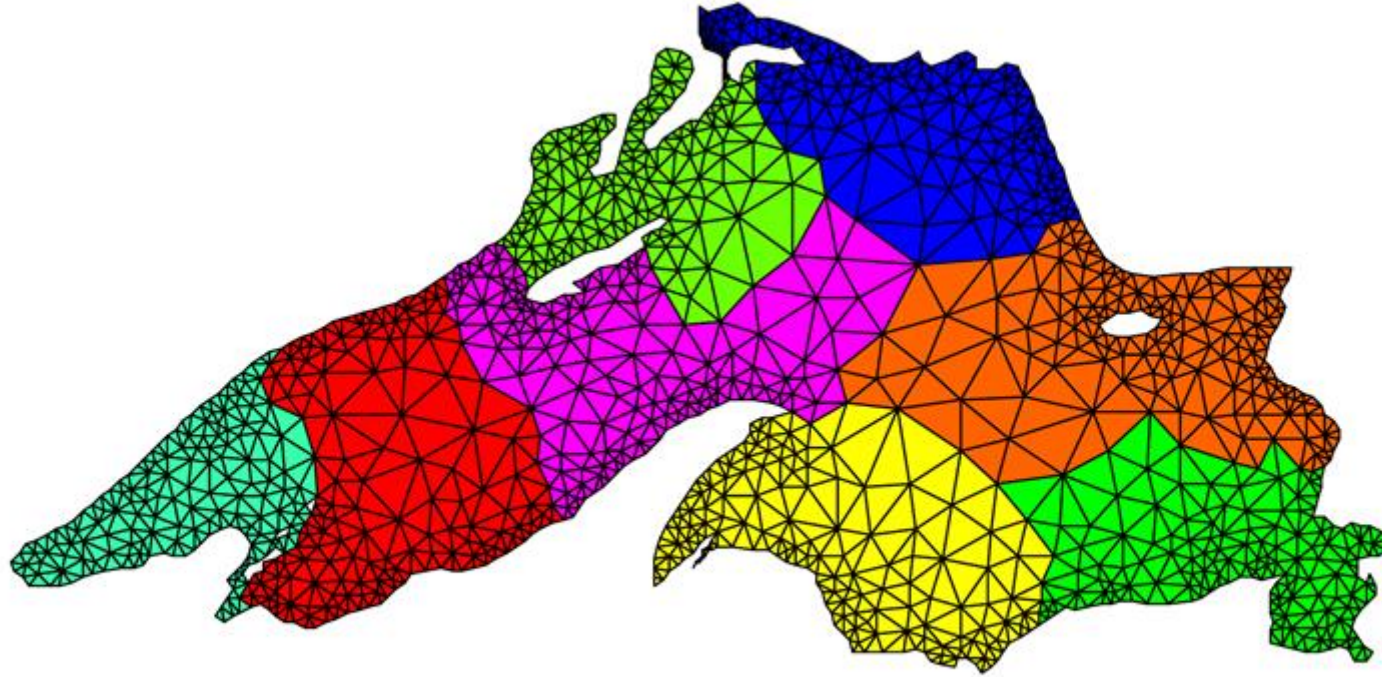


Random distribution



A random distribution of the mesh elements to eight processes

Graph Partitioning



A distribution of the mesh elements to eight processes, by using a graph-partitioning algorithm

Summary

- **Mapping Schemes**

- **Static Mapping**

- ✓ Tasks distribution among the processes is done before the execution of the program
 - ✓ Total number of tasks and their sizes are known a priori

- **Dynamic Mapping**

- ✓ Tasks distribution among the processes is done during the execution of the program
 - ✓ Total number of tasks and their sizes are not known a priori

- **Schemes for Static Mapping**

- **1D & 2D Block distribution**

- **Cyclic distribution, Block-Cyclic distribution**

- **1D & 2D Randomized-Block distribution**

Summary (Cont.)

- **Randomized block cyclic distribution is not always used**
 - **Poor spatial locality**
 - **High inter-process interaction**
- **Random Distribution**
 - **A simulation model (using a mesh of tasks) for finding dispersion of water contaminant**
- **Graph Partitioning**
 - **A simulation model (using a mesh of tasks) for finding dispersion of water contaminant**

Additional Resources

- **Introduction to Parallel Computing** by **Ananth Grama** and **Anshul Gupta**

➤ **Chapter 3. Principles of Parallel Algorithm Design**

✓ **Section 3.4: Mapping Techniques for Load Balancing**

Questions?